Beyond the Wug: Unmasking the Reliance on Distributional Semantics for Deductive Reasoning in Large Language Models

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Abstract

A central debate in the cognitive science of artificial intelligence is whether Large Language Models (LLMs) possess true formal reasoning capabilities or if they simulate deductive logic through advanced statistical pattern matching. This paper evaluates the deductive reasoning capabilities of three distinct LLM architectures (GPT-4o-Mini, GPT-5 Mini, and Gemini Flash Lite) by contrasting their performance on standard English syllogisms against syntactically identical but semantically scrambled (out-of-vocabulary) syllogisms. While all models achieve near-perfect accuracy on standard logic, their performance degrades significantly when semantic priors are removed, exposing a heavy reliance on distributional semantics. Furthermore, we demonstrate that explicit Chain of Thought (CoT) prompting acts as a compensatory cognitive scratchpad for semantically dependent models, but introduces an "overthinking penalty" for models with already robust syntactic parsing. These findings suggest that true advances in artificial reasoning require architectural shifts rather than heuristic prompt engineering.

1 Introduction

The appearance of logical reasoning in modern Large Language Models (LLMs) often masks the underlying mechanisms driving their outputs. When an LLM correctly deduces that a cat is an animal because all mammals are animals, it is difficult to isolate whether the model is traversing a formal logical tree (syntactic parsing) or simply predicting highly probable tokens based on word co-occurrences in its training data (distributional semantics).

This study tests the hypothesis that LLMs rely disproportionately on semantic priors rather than formal logical structures. By replacing familiar nouns with out-of-vocabulary (OOV) pseudo-words (e.g., "wugs," "florps"), we zero out semantic weights, forcing the models to rely entirely on syntactic operators. Additionally, we investigate whether explicit Chain of Thought (CoT) prompting bridges this syntactic gap by forcing symbolic variable mapping.

2 Experimental Methodology

2.1 Model Selection

To ensure a robust architectural comparison, three models were evaluated:

- **GPT-4o-Mini:** A baseline for highly optimized, widespread autoregressive transformers.
- **GPT-5 Mini:** Selected to evaluate whether latent, hidden reasoning architectures inherently process abstract logic better than their predecessors.
- **Gemini Flash Lite:** Included to provide a cross-industry contrast, testing Google's distinct attention and routing mechanisms.

2.2 Dataset Design

The evaluation pipeline utilized 24 logical syllogisms (19 formally valid, 5 formal fallacies) evaluated under two conditions:

- **Condition A (Standard):** Syllogisms utilizing highly probable English nouns.
- **Condition B (Scrambled):** Syllogisms preserving the exact logical structure but substituting substantive tokens with semantic nonsense.

2.3 Prompting Strategies

Each model evaluated the dataset under two prompting constraints:

1. **Zero-Shot Direct:** Models output only a binary "VALID" or "INVALID" classification.

2. **Explicit Chain of Thought (CoT):** Models were instructed to map premises to abstract variables (A, B, C) step-by-step prior to outputting a final classification.

3 Results and Observations

3.1 The Baseline Illusion

Under the Zero-Shot Standard condition, all three models achieved an identical accuracy of **95.8%**. In isolation, this benchmark suggests a mastery of deductive logic. However, this performance is heavily inflated by distributional semantics; the models predict logical validity based on the rich parametric memory of the entities involved rather than the strict application of logical rules.

3.2 The Syntactic Breakdown

Isolating the logical operators via OOV tokens (Condition B) revealed stark architectural vulnerabilities.

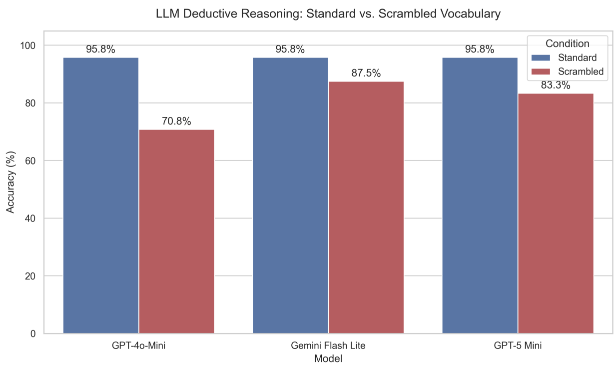

Figure 1: Baseline accuracy comparison demonstrating the performance drop when semantic priors are removed.

As shown in Figure 1, removing semantic anchors fractured the models' ability to parse logic:

- **GPT-4o-Mini:** Accuracy plummeted to **70.8%**. This severe degradation confirms its reasoning is highly semantically dependent.
- **GPT-5 Mini:** Accuracy dropped to **83.3%**. The latent reasoning architecture halved the error rate of its predecessor.
- **Gemini Flash Lite:** Accuracy dropped to **87.5%**. Gemini proved to be the most syntactically robust in a zero-shot environment.

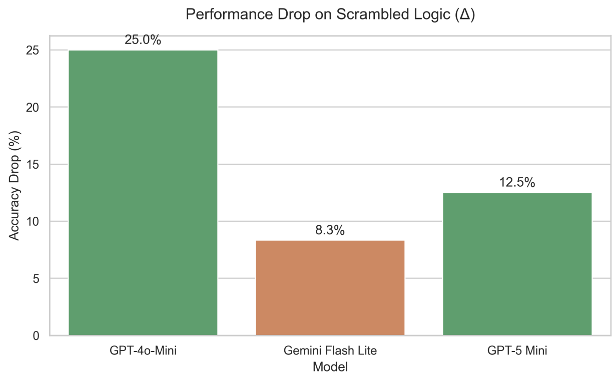

Figure 2: Absolute performance drop (Δ) representing the semantic subsidy required by each architecture.

3.3 The Chain of Thought Paradox

The application of explicit CoT prompting yielded the most revealing data regarding how different architectures handle abstract symbolic logic (Figure 3).

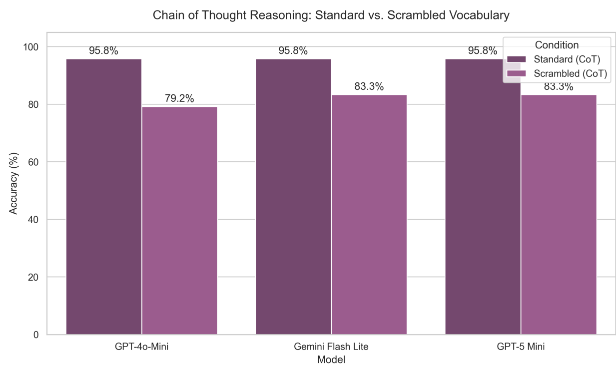

Figure 3: Model accuracy when utilizing explicit Chain of Thought prompting.

When models were forced to "think step by step," the outcomes diverged based on inherent baseline robustness:

- **The Rescue Effect:** GPT-4o-Mini experienced a gain of **+8.4%** on scrambled data. For the weakest syntactic parser, forcing a step-by-step variable mapping created a cognitive "scratchpad" that compensated for missing semantic anchors.
- **The Overthinking Penalty:** Conversely, Gemini Flash Lite suffered a **-4.2%** penalty, and GPT-5 Mini saw no gain. For models with robust internal syntactic parsing, explicitly narrating symbolic logic using nonsense words introduced noise, degrading performance.

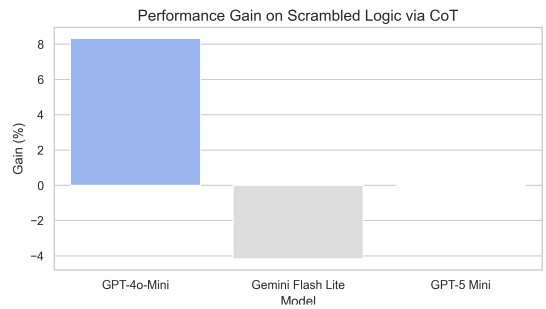

Figure 4: Net accuracy gain or loss when utilizing CoT on scrambled syllogisms.

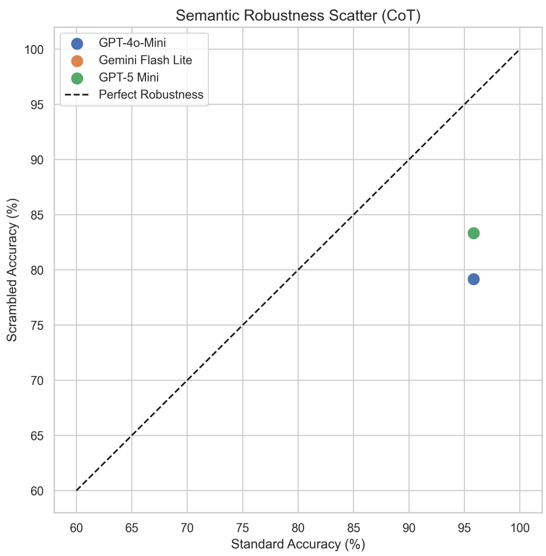

Figure 5: Semantic Robustness Scatter showing model reliance on semantics vs. syntactic structure.

4 Conclusion

This evaluation provides concrete empirical evidence that standard LLM deductive reasoning is heavily subsidized by distributional semantics. When familiar semantic priors are removed, logical capabilities fracture.

Crucially, the data suggests a definitive ceiling to prompt engineering. While explicit Chain of Thought helps weaker, semantically-reliant models bridge the syntax-semantics gap, it acts as a disruptive cognitive load for structurally robust models when processing abstract variables. True advances in artificial reasoning require fundamental architectural shifts—such as the integration of latent reasoning spaces—rather than reliance on explicit, user-guided cognitive scratchpads.